Dr. Gao Zuchang

Senior Scientist & Manager

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Research Profile

Applied researcher and R&D leader specializing in **intelligent lithium battery management systems**, **fuel cell control technology**, and hybrid **energy solutions**. Over 10 years of experience bridging academic research and industrial product development, with a proven track record of translating laboratory innovations into commercially deployed products. PI and technical lead on research projects totaling over **SGD 4 million** in competitive grants and over **RMB 45 million** in international industry partnerships. 25 publications with 860+ citations.

Education

Ph.D., Intelligent Battery Management and Fault Diagnosis 2013 – 2017
Newcastle University, Newcastle upon Tyne, UK
Research focus: $LiFePO_4$ battery modelling, state estimation, fault diagnosis

International Exchange Student, Fuel Cell Characteristics and Control Systems 2008 – 2009
Temasek Polytechnic, Singapore

M.Eng, Electrical Engineering and Automation (Power Electronics) 2007 – 2010
Southwest Jiaotong University, Chengdu, China

B.Eng, Electrical Engineering and Automation 2003 – 2007
Southwest Jiaotong University, Chengdu, China

Professional Experience

Senior Scientist & Manager 2017 – Present
Clean Energy Research Centre (CERC), Temasek Polytechnic, Singapore

- Lead the Power System Control division covering **lithium battery systems** and **fuel cell systems**; manage a team of **5 engineers and scientists**.
- Serve as **PI/Co-PI** on multiple national and industry-funded research projects totaling over **SGD 4M** in grants.
- Spearheaded development of **DUROXO smart lithium battery solution for forklifts** – from concept through VDE certification (IEC 62619) to commercial launch across **7 countries**. Product sales over **SGD 8 Million** as of Mar 2026.
- PI for Singapore's first fuel cell-powered long-endurance drone (NRF grant, SGD 1.2M), achieving >**5x flight endurance** vs commercial pure battery solution.
- **Technical lead** for the CQFC joint venture (RMB 45M), directing fuel cell control system R&D and commercialization.
- Manage international research collaborations with Southwest Jiaotong University, Chongqing University, and Wuhan University of Technology (initiator and coordinator for MoU signing in 2024).

Engineer 2009 – 2013

Clean Energy Research Centre (CERC), Temasek Polytechnic, Singapore

- Conducted R&D on fuel cell characteristics, control strategies, and battery power system design for marine and EV applications.
- Contributed to teaching support including student laboratory supervision and project guidance.

Research Grants & Funded Projects

*Total funding secured as PI/co-PI: ~**SGD 4 million + RMB 45 million** (international partnership)*

Period	Project Title	Role	Funder	Amount
2019 – 2023	Long Flight Endurance Drone with AI/Analytics-Enabled Detection	PI	NRF CRP	SGD 1.2M
2020 – Now	Fuel Cell International Cooperation – CQFC Joint Venture	Tech Lead	Industry (CN)	RMB 45M
2019 – 2022	Configurable Smart LFP Battery System for MHE (DUROXO)	PI	Industry+ESG	SGD 560k
2020 – 2022	Next-Gen Li-Ion ESS and Smart Distributed ESS Management	Co-PI	Industry	SGD 250k
2023 – 2025	Multi-Stack Control & Energy Hybrid for Air-Cooled H_2 PEM FC	PI	MOE	SGD 178k
2023 – 2025	Hot-Swappable Smart Li Battery BMS for 2-3 Wheeled Vehicles	PI	MOE	SGD 162k
2023 – 2024	Portable PEM FC Powered Battery Charger	PI	Industry	SGD 30k
2017 – 2019	High Performance Li-Ion Battery for Deep Water Operation	Co-PI	EMA/SMI	SGD 680k
2023 – 2025	Smart IoT Kit with Battery Temp. Mgmt. for Subzero Env.	Co-PI	MOE-TIF	SGD 395k
2023 – 2025	Modular Intelligent Lightweight Lithium Battery Charger	Co-PI	MOE-TPRF	SGD 171k
2024	BMS Integration for Wireless Charging in Autonomous Forklifts	PI	Industry	SGD 10k
2025 – 2026	Applied AI: Telemetry System for Universal Application	PI	Industry	SGD 20k
2025 – 2026	PiezoBMS: Piezoelectric Tech. for Battery Pack Performance	PI	MOE-TPRF	SGD 120k
2025 – 2026	Eco-BMS: Battery Mgmt. Solution for Material Handling	PI	Industry	SGD 13k
2026 – 2028	Battery Safety: Online EIS for Health Diagnostics	PI	MOE-TPRF	SGD 196k

Industry Impact & Technology Transfer

Commercial Products (PI)

- **DUROXO Lithium-Ion Solutions** – Smart $LiFePO_4$ BMS for forklifts featuring active balancing, SOC/SOH estimation, fault diagnostics, 4G/WiFi IoT, and web-based monitoring. VDE-certified (IEC 62619:2017). Deployed in **7 countries** across Asia. Featured by Channel NewsAsia (Aug 2020). Product sales over **SGD 8 Million** as of Mar 2026.

- **Portable H_2 Fuel Cell Tower Light** – Collaboration with Powerzone H2 Technologies for construction/mining sector. **30–40%** higher power density vs. conventional Li battery solutions. Product launched Jan 2025. Secured **first 100 sets** of order from Indonesia mining customer.

- **Universal Telemetry Platform** – Partnership with local fuel cell leading manufacturer Spectronik Technology for universal IoT telemetry platform to help any traditional equipment for power

monitoring and diagnosis. Product launched Mar 2026 (including Amazon US). Secured first order within one week after launched.

- **Singapore's First Fuel Cell Drone** – H_2 fuel cell-powered long-endurance drone (25 kg class) with AI-enabled inspection. Deployed for PUB smart water pipeline inspection (Mar 2023), achieving **>2 hr** flight endurance (**5x** vs traditional pure battery power).Commercial discussion and verification process in progress with overseas customer.

International Technology Partnerships

- **CQFC Joint Venture** – Technical lead for a Singapore-China JV on fuel cell stack and control system commercialization (est. Apr 2020, RMB 45M).
- **TP–WUT Research Collaboration** – Initiated and coordinate research collaboration and student exchange with Wuhan University of Technology; MoU signed Jun 2024.

Awards & Honours

- **Public Sector Transformation Award 2023** – Dare to Do (PI), Singapore Government, for the DUROXO Smart Lithium Battery System.
- **TP Exemplary Overcomer Award 2023** – Recognized as PI for the Lithium Battery Project.
- **Impact Award, TP RIE Awards 2025** – For contributions to PEM Fuel Cell Stack Technology.
- **Shell Eco-Marathon 2026 (Qatar)** – TP Eco Flash EE control Team Lead; Won the 2nd place among Asia, Urban Concept Hydrogen Fuel Cell category.

Teaching & Mentoring

- **Final Year Project Supervision** – Regularly supervise diploma students on industry-relevant projects in BMS, fuel cell control, and IoT-enabled energy systems at Temasek Polytechnic.
- **International Exchange Student Mentoring** – Coordinate and supervise exchange **undergraduate and postgraduate** students from Wuhan University of Technology (WUT), Chongqing University (CQU), and Southwest Jiaotong University (SWJTU) on research and industry-focused projects.
- **Continuing Education & Training (CET)** – Deliver professional training programs:
  - Industry Best Practices in Compliance with International Safety Standards for Safe Installation, Operation & Maintenance of Energy Storage Systems (SP Group)
  - Introduction to Energy Storage Systems (SP Group)
  - Fuel cell stack and control systems (Industry partners)
  - Lithium Battery Pack Upgrade & Maintenance Training for Extended Drone Flight Range (Power & Energy Engineering Centre, HQ MES/SAF)

Selected Publications

Full list: 25 publications | 860+ citations | Google Scholar Profile

[1] **Z. Gao**, C.S. Chin, W.L. Woo, J. Jia, "Integrated Equivalent Circuit and Thermal Model for Simulation of Temperature-Dependent $LiFePO_4$ Battery in Actual Embedded Application," *Energies*,

10(1), 85 (2017).

[2] J. Jia, G. Wang, **Z. Gao**, M. Han, "Experimental Study on CNT Heating for Li-Ion Batteries in Extremely Low-Temperature Environments," *Energies*, 18(11), 2958 (2025).

[3] B. Xiong, F. Wei, Y. Wang, K. Xia, F. Su, Y. Fang, **Z. Gao**, Z. Wei, "Hybrid Robust-Stochastic Optimal Scheduling for Multi-Objective Home Energy Management," *Energy*, 290, 130047 (2024).

[4] Z. Fu, L. Lu, C. Zhang, Q. Xu, X. Zhang, **Z. Gao**, J. Li, "Fuel Cell and Hydrogen in Maritime Application: A Review," *Sust. Energy Technol. Assess.*, 57, 103181 (2023).

[5] B. Zhang, Y. Feng, C. Zhang, **Z. Gao**, et al., "Modeling, Control and Analysis of a Novel Energy-Saving Fuel Cell Anode System," *Energy Conv. Mgmt.*, 283, 116942 (2023).

[6] C.S. Chin, C. Zhang, **Z. Gao**, "Deploying Battery Technology for Marine Vessel Electrification," *IEEE Potentials*, 40(6), 24–33 (2021).

[7] C.S. Chin, **Z. Gao**, M. Han, C. Zhang, "Enhancing Performance of PV Panel by Cold Plate Design with Guided Channels," *IET Renew. Power Gen.*, 14(9), 1606–1617 (2020).

[8] C.S. Chin, C. Zhang, **Z. Gao**, "Comprehensive Electro-Thermal Model of 26650 Lithium Battery for Discharge Cycle," *J. Energy Storage*, 28, 101222 (2020).

[9] C.S. Chin, **Z. Gao**, "Thermal State-of-Expansion or Melting of PCM Based Heat Sink for Underwater Battery Power System," *J. Energy Storage*, 26, 100956 (2019).

[10] C.S. Chin, **Z. Gao**, "System Design of Underwater Battery Power System for Marine and Offshore Industry," *J. Energy Storage*, 21, 724–740 (2019).

[11] C.S. Chin, **Z. Gao**, C. Zhang, "A Pseudo 3D Electrochemical-Thermal Model of a Cylindrical $LiFePO_4$/Graphite Battery," *Appl. Therm. Eng.*, 147, 450–463 (2018).

[12] L. Xia, C. Zhang, M. Hu, S. Jiang, C.S. Chin, **Z. Gao**, Q. Liao, "Investigation of Parameter Effects on HT-PEM Fuel Cell Performance," *Int. J. Hydrogen Energy*, 43(52), 23441–23449 (2018).

[13] C.S. Chin, **Z. Gao**, C. Zhang, "Nonlinear Temperature-Dependent State Model of Cylindrical $LiFePO_4$ Battery for SOC Estimation with EKF," *Energies*, 11(9), 2467 (2018).

[14] C.S. Chin, **Z. Gao**, "Low-Temperature Macro-Encapsulated PCM Based Thermal Energy Storage System," *Appl. Therm. Eng.*, 141, 928–938 (2018).

[15] C.S. Chin, **Z. Gao**, "SOC Estimation of Battery Pack under Varying Ambient Temperature Using Adaptive Sequential ELM," *Energies*, 11(4), 711 (2018).

[16] **Z. Gao**, C.S. Chin, W.L. Woo, J. Jia, "SOC Estimation and Active Cell Pack Balancing Design of Li Battery Power System for Smart EV," *J. Adv. Transport.*, 4028732 (2017).

[17] J. Jia, **Z. Gao**, B.C.C. Peng, H.C. Looi, "Development of a 3kWh Battery-Supercapacitor Hybrid Energy Storage System," *IEEE TENCON 2024*, pp. 1901–1904 (2024).

[18] **Z. Gao**, J. Jia, et al., "Modelling and Simulation of a 12-Cell Battery Power System with Fault Control for Underwater Robot," *IEEE 7th Int. Conf. CIS* (2015).

[19] **Z. Gao**, J. Jia, et al., "GA-Based BPNN Approach for Fault Diagnosis in Lithium-Ion Battery System," *6th Int. Conf. PESA* (2015).

[20] P. Krishnamoorthy, C.S. Chin, **Z. Gao**, W. Lin, "A Multi-Hop Microprocessor Based Prototype System for Remote Vibration and Image Monitoring," *IEEE 7th Int. Conf. CIS* (2015).

Patents

[1] Phase Change Material Based Heat Sink with Volumetric Expansion Control – Singapore Patent Application.

[2] "A fuel cell and hydrogen tank installation structure for UAV" (ZL 2023 2 2435111.2), granted Apr 2024.

[3] "A method to improve fuel cell performance through a short-circuit approach" (Filed, CN, under review).

[4] "A method for controlling output voltage range of air-cooled FC system based on circuit switching" (Filed, CN, under review).

[5] "A device with integrated fuel cell controller interface and heat dissipation" (Filed, CN, under review).

[6] "An exhaust control strategy for air-cooled PEM fuel cells" (Filed, CN, under review).

Research Interests

- Li-ion battery modelling, SOC/SOH estimation, fault diagnosis, and smart BMS
- PEM fuel cell characteristics, control, thermal management, multi-stack hybridization
- Power hybridization for ESS, EVs, marine vessels, UAVs, and MHE
- Smart energy IoT: embedded systems, wireless BMS, cloud diagnostics, edge computing
- Electrochemical impedance spectroscopy (EIS) for battery diagnostics

Technical Expertise

Embedded Systems: Microchip/NXP/STM32/ESP32, FreeRTOS, UART/SPI/I2C/CAN, BMS firmware development

Hardware Design: PCB design (KiCad, Altium Designer), power electronics, analog/digital circuit, DC-DC converters

Software & Simulation: MATLAB/Simulink, .NET (IoT web apps), Python, C/C++

Domain Expertise: $LiFePO_4$/NMC battery systems, PEM fuel cell systems, active cell balancing, EIS, thermal management, VDE/IEC certification

Professional Service

Singapore Standard Publication Involvement

- SS 716:2025 – Railway applications: Track-safety protection during work; competences for personnel related to work on or near tracks.
- SS 717:2025 – Obsolescence management: Railway.

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